

Ring resonator-based dynamic controller for precise wavelength separation of a DWDM laser source

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Abstract: We demonstrate a ring-resonator based controller that provides simultaneous and continuous control of the wavelength spacing of a distributed feedback laser array maintaining it within 201 ± 4 GHz in the presence of ambient variations.

1. Introduction

Recently, dense wavelength division multiplexing (DWDM) for optical transceivers (OTRX) that dramatically increases the per-fiber data rate of co-packaged silicon photonic interconnects has been shown [1-3]. Uniform power and precise spacing ($\Delta\lambda$) are critical for multi-wavelength laser (MWL) sources in DWDM. A 16- λ Distributed Feedback (DFB) laser array spaced at 200 GHz ($\Delta\lambda$) with ~ 16 GHz spacing error ($\sigma_{\text{std-dev}}$) was shown in [4] and is compliant with the CW-WDM multi-source agreement specifying < 50 GHz grid spacing error [5]. However, such several tens of GHz of error between adjacent carriers will still limit the data rate per- λ due to channel crosstalk [6,7]. Another potential advantage of lower error in $\Delta\lambda$ is simplifying the alignment of ring resonator-based receiver-side demultiplexers [3,7]. If the incoming DWDM signal is on a fixed $\Delta\lambda$ -grid, the periodic demux array can be collectively tuned [7], potentially reducing the complexity and tuning time of the ring thermal control logic.

Mode-locked and comb lasers [8,9] can enable precise $\Delta\lambda$ defined by a single cavity, but have lower output power and greater power variation across their spectrum than DFB arrays. In this work, we demonstrate an electro-optic controller (Fig. 1) which aligns and actively locks the DFB array we presented in [10] to a 201 ± 4 GHz $\Delta\lambda$ -grid defined by the free-spectral range (FSR) of a micro-ring resonator (MRR). The resulting MWL source provides the power uniformity of a DFB array with $< 2\%$ spacing error without sub-nm precision in fabrication. Any residual power non-uniformity in the array can be corrected on a per- λ basis with a dedicated integrated optical amplifier [11] or variable attenuator.

An important feature of the proposed control scheme is to simultaneously and continuously control the λ of all the lasers in the array in the presence of external variations. To distinguish between lasers with all simultaneously incident on the same ring resonator, each laser is modulated (marked) with a unique tone to identify and continuously correct any misalignment from the desired $\Delta\lambda$ -grid.

2. Wavelength-spacing controller description

The eight laser outputs are combined and routed to the ring-based controller. A $60 \mu\text{m}$ radius MRR with 200 GHz FSR fabricable with ± 3 GHz accuracy (due to its large feature size) defines a temperature invariant $\Delta\lambda$ -grid. The controller objective is to align λ_{1-8} of all eight lasers (L_{1-8}) to consecutive ring resonant peaks.

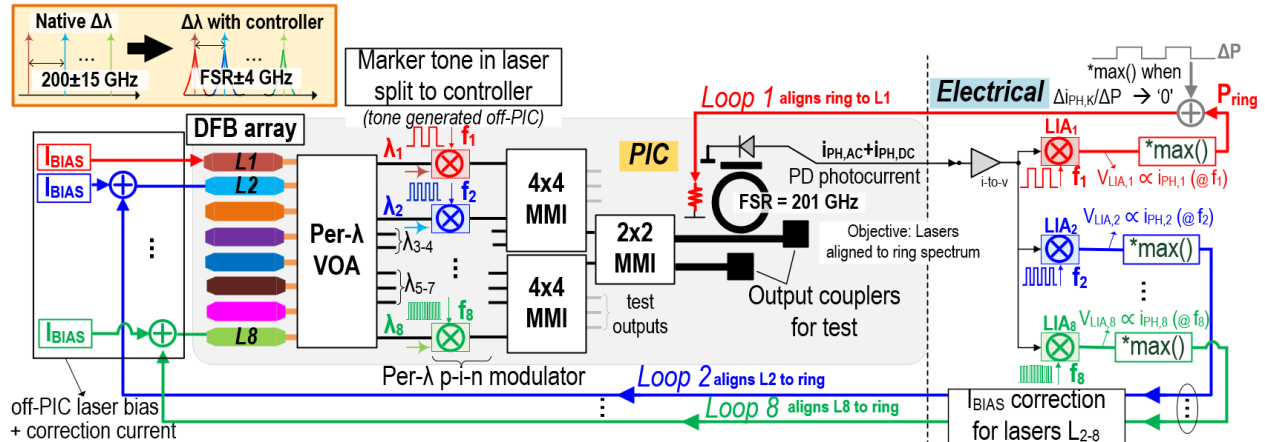


Fig. 1. Block diagram of ring resonator-based wavelength-spacing controller for multi- λ DFB array.

First, one of the MRR peaks is aligned to laser L_1 by tuning the ring heater as shown in Fig. 2(a). Then the other lasers ($\lambda_{k=2-8}$) are locked to the resonant peak closest to their λ_k by tuning the respective laser bias current. Given less than tens of GHz of native spacing error, each laser will lie within the capture range of the unique and consecutive MRR peaks. This controller does not define the absolute lasing λ_k for any laser, but by locking adjacent lasers to consecutive ring resonances it reduces the error in $\Delta\lambda$ of the DFB array in [10] to $<2\%$.

If a single laser is aligned to an MRR peak, the power absorbed in the MRR and the photocurrent (i_{PD}) at the drop port Ge photodiode (PD) is maximized. However, with all lasers incident on the ring, maximizing the total dc photocurrent ($i_{PD,DC}$) does not indicate an individual laser's misalignment from the desired $\Delta\lambda$ -grid. To distinguish among the wavelengths, each laser L_k is intensity modulated with a unique square wave of frequency f_k , by an in-line Si p-i-n junction modulator in the laser output to the controller (Fig. 1).

In Fig. 1, all of the laser power is sent to the controller. However, to use this controller with a DWDM OTRX, the laser outputs would be split between the controller ($<10\%$ power) and the desired application (e.g. $>90\%$ power to the 8- λ OTRX). By not modulating the lasers directly, the power split ($>90\%$) to the OTRX is marker tone free. Fig. 2(b) shows the photonic IC (PIC) die with the laser array, p-i-n modulators and 60 μm radius MRR. With all the laser power being routed to the MRR, we have included a per- λ variable optical attenuator (VOA) in the path to the controller and used this to equalize the laser power after the lasers are locked to the desired $\Delta\lambda$ -grid while simultaneously mitigating ring self-heating. Semiconductor optical amplifiers (SOA) could instead be integrated [11] in the ($>90\%$) path to the OTRX in future implementations to further equalize laser power.

The drop-port PD current consists of the dc current $i_{PD,DC}$, and ac photocurrent ($i_{PD,AC}$) from the intensity modulated lasers. $i_{PD,AC}$ has content at the fundamental tones f_k (noted as $i_{PH,K}$) and their harmonics, as well as mixing terms due to PD and other nonlinearities. The tones f_k are planned to ensure no overlap of harmonics or intermodulation terms ($\leq 3^{\text{rd}}$ order) with any fundamental, and the fundamental component $i_{PH,K}$ is a unique indicator of the detuning ϵ_K of laser L_K from its closest resonance λ_{0K} ($\epsilon_K = \lambda_K - \lambda_{0K}$) (Fig. 2(c)). Each $i_{PH,K}$ can be detected by digitizing the photocurrent $i_{PD,AC}$ and analyzing its spectral content, or by using a dedicated lock-in amplifier (LIA_K in Fig. 1). The wavelength error ϵ_K for each laser can therefore be monitored continuously using the same MRR.

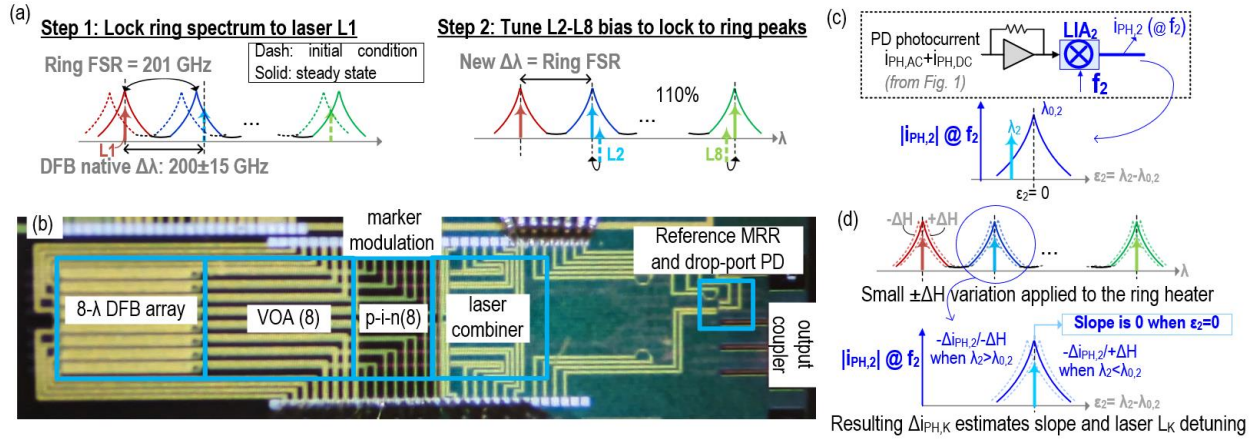


Fig. 2. (a) 200±15 GHz spaced DFB array is tuned to align lasers with ring resonances, (b) PIC with optical components for λ -spacing controller, (c) $i_{PH,K}$ at f_k measures laser L_K detuning from ring (ϵ_K) (shown for $K = 2$), (d) slope-based alignment to ring peaks.

To align any laser to the closest ring resonance using its corresponding $i_{PH,K}$, we employ slope-minimization as shown in Fig. 2(d). Based on the detuning of laser L_k from the ring peak (ϵ_K), the amplitude of $i_{PH,K}$ at f_k follows the MRR Lorentzian absorption spectrum closely. The slope at an absorption peak is zero. Therefore, a non-zero slope is a metric for ϵ_K , where the sign indicates either $\lambda_K < \lambda_{0K}$, or $\lambda_K > \lambda_{0K}$.

Slope-minimization enables laser λ alignment without a-priori knowledge of the laser power, and allows the laser bias current to be used as a laser wavelength control knob. For the DFB array in [10], correcting the $\Delta\lambda$ -grid error can require a large change in the bias current with ~ 3 dB power variation, and aligning the laser to the ring by locking to an absolute i_{PH} is not viable. Our slope-based locking method and algorithm are agnostic to power variation from laser current bias tuning.

3. Measured Results

We demonstrate the proposed $\Delta\lambda$ -controller with the PIC wirebonded to a PCB and external equipment to provide bias and control (Fig. 1). No TEC or heat sinks are included in the experiment. The laser on the PIC is an 8- λ III-V/Si hybrid DFB array [10] biased at 50 mA initially with 200±15 GHz spacing. The measured FSR of the 60

μm radius ring is 201 GHz, and is within the ± 3 GHz expected spread due to fabrication. Eight marker tones (f_k) at 25.3, 29.1, 31.2, 33.5, 37, 39.7, 44, 46.5 kHz modulate lasers 1-8 respectively. The control loop is closed using external lock-in amplifiers and the loop bandwidth is low. This bandwidth can be expected to improve significantly when paired with a CMOS based electronic IC for control.

To estimate the slope, a square wave of ± 0.5 mW (ΔH) is applied to the ring heater (Fig. 2(d)) and is limited by the 0.2 mA resolution of the external current module. The resulting change in the photocurrent $i_{\text{PH},K}$ at f_k (from laser L_K) is recorded from the output voltage change ($\Delta V_{\text{LIA},K}$) of the corresponding LIA. For this measurement, laser L_5 in the center of the array is used as reference and its current stays at 50 mA. Based on $\Delta V_{\text{LIA},5}$, the current to the ring heater is adjusted to drive the slope of $V_{\text{LIA},5}$ (or $i_{\text{PH},5}$) to zero, aligning an MRR absorption peak to laser L_5 . The other LIA outputs simultaneously adjust the bias current of lasers $L_{1-4, 6-8}$ to also drive the slopes of their LIA outputs to '0'. The MRR heater current and laser bias are shown in Fig. 3(a). The relative spacing of lasers 1-8 are shown for the same time span in Fig. 3(b). Figs. 3(c) and (d) show the free-running spectrum and the spectrum at steady state with the laser array aligned to a 201 GHz ring FSR with ± 4 GHz error due to finite loop gain and tracking bandwidth. The free running DFB array has < 2 dB power variation without any VOA equalization. The per- λ VOA is adjusted to equalize the power of the $\Delta\lambda$ -locked DFB array to achieve < 2 dB variation as well.

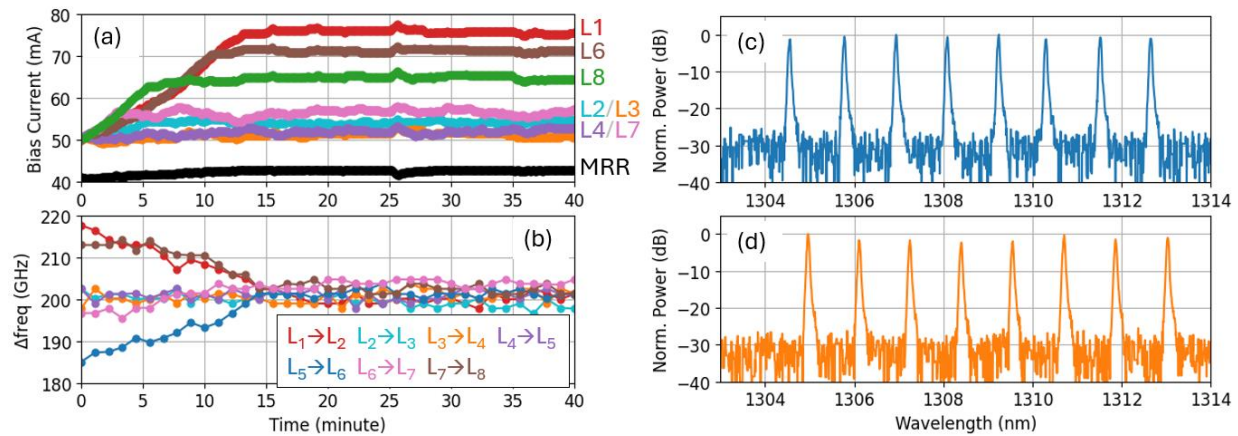


Fig. 3(a) Evolution of heater and laser bias as the controller locks the DFB array to a $\Delta\lambda$ -grid defined by the ring FSR, (b) frequency spacing of the 8-laser array, and (c) initial free-running spectrum and steady state spectrum at 40 min, both with < 2 dB power variation.

We have demonstrated a laser wavelength spacing controller for DWDM applications that aligns an 8- λ DFB array with 200 ± 15 GHz nominal spacing to the 201 ± 4 GHz $\Delta\lambda$ -grid defined by the temperature-invariant FSR of a $60 \mu\text{m}$ radius MRR. We modulated each laser with a unique frequency, so the misalignment is uniquely identifiable when all lasers are incident on a single ring. Our approach eliminates any need for time-interleaved control when using a ring resonator with multiple lasers simultaneously, and enables continuous control of all lasers in the presence of ambient variations.

4. Acknowledgement

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5. References

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